Lecture 11: The Physics of Fission

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

February 6, 2026

1 Introduction: The Destination

In Lecture 10, we discussed the "Journey" of the neutron: slowing down from 2 MeV to 0.025 eV, avoiding resonance traps, and thermalizing. Today, we discuss the "Destination": **Fission**.

What exactly happens when a thermal neutron strikes a uranium nucleus? How does the nucleus split? And most importantly for engineers, where specifically does the energy come from?

2 The Liquid Drop Model (Bohr & Wheeler, 1939)

To understand the mechanics of fission, we model the heavy nucleus not as a collection of hard spheres, but as a drop of incompressible, charged fluid. The stability of this drop is determined by a tug-of-war between two fundamental forces:

1. **Nuclear Force (Surface Tension):** The strong nuclear force acts between nearest neighbors. Just like surface tension in a water drop, this force tries to keep the nucleus spherical (minimizing surface area).
2. **Coulomb Force (Repulsion):** The nucleus contains 92 protons (for Uranium) all repelling each other. This force tries to tear the nucleus apart.

2.1 The Activation Energy

In a stable nucleus like ^{235}U (well, almost stable anyway), the surface tension is just strong enough to contain the Coulomb repulsion. However, the equilibrium is fragile.

- When a neutron is absorbed, it adds **Binding Energy** ($\sim$6 MeV) and (if unmoderated) kinetic energy to the system.
- This energy excites the nucleus, causing it to vibrate and oscillate.
- If the oscillation deforms the sphere into a "dumbbell" or peanut shape, the Coulomb distance between the two lobes increases, but the surface tension (neck area) decreases.
- **Critical Deformation:** Once the nucleus stretches past a certain point, the Coulomb repulsion between the two lobes overcomes the restoring force of the surface tension. The split becomes inevitable.

3 The Dynamics of "The Snap"

The actual moment of separation is violent and complex. As Chemical Engineers, we can visualize this using a familiar fluid dynamics analogy.

3.1 The "Satellite Droplet" Analogy

Consider the breakup of a viscous liquid jet or a dripping faucet. As the drop stretches, a thin fluid "neck" forms between the two main lobes. Due to the Rayleigh-Plateau instability, this neck eventually snaps.

1. **In Fluid Mechanics:** When the neck breaks, the fluid contained within it does not simply vanish. It often breaks up into tiny, secondary droplets known as **Satellite Droplets** that are found between the two main drops.

2. **In Nuclear Fission:** The "nuclear fluid" in the neck behaves identically. When the neck snaps, the nucleons in that region are left stranded. They are ejected immediately as free neutrons.

This mechanism gives rise to the **Prompt Neutrons** (ν) that sustain our chain reaction. They come from two distinct sources during the snap:

- **Scission Neutrons (∼10%):** These are the "satellite droplets" ejected directly from the snapping neck at the moment of separation ($t \approx 10^{-20}$ s).

- **Evaporation Neutrons (∼90%):** Immediately after the snap, the two large fragments are highly deformed and incredibly "hot." As they relax into spherical shapes, they "boil off" excess neutrons to cool down ($t \approx 10^{-17}$ s).

Result: On average, the fission of ^{235}U releases $\nu \approx 2.43$ neutrons.

4 The Semi-Empirical Mass Formula (SEMF) *(Lamarsh, 2.12)*

To understand why some isotopes fission and others do not, we must quantify the nuclear binding energy. We use the **Semi-Empirical Mass Formula (SEMF)**, based on the Liquid Drop Model proposed by George Gamow and refined by Weizsäcker.

The binding energy $B(A, Z)$ is approximated by five distinct physical terms:

$$B(A,Z) = \underbrace{a_v A}_{\text{Volume}} - \underbrace{a_s A^{2/3}}_{\text{Surface}} - \underbrace{a_c \frac{Z(Z-1)}{A^{1/3}}}_{\text{Coulomb}} - \underbrace{a_{sym} \frac{(N-Z)^2}{A}}_{\text{Asymmetry}} + \underbrace{\delta(A,Z)}_{\text{Pairing}} \tag{1}$$

Physical Interpretation of the Terms

1. **Volume Term ($a_v A$):** The strong nuclear force is short-range and saturates. Each nucleon interacts only with its nearest neighbors. Thus, the binding energy is directly proportional to the volume (or total mass number A).

2. **Surface Term ($-a_s A^{2/3}$):** Nucleons on the surface of the "drop" have fewer neighbors than those in the center. This reduces the total binding energy. This effect is proportional to the surface area ($4\pi R^2 \propto A^{2/3}$).

3. **Coulomb Term ($-a_c Z^2/A^{1/3}$):** The protons repel each other due to the electromagnetic force. This instability grows with the square of the proton number (Z^2) and decreases as the nucleus gets larger (inversely proportional to radius $A^{1/3}$).

4. **Asymmetry Term ($-a_{sym}(N-Z)^2/A$):** Due to the Pauli Exclusion Principle, the nucleus is most stable when $N \approx Z$. Having a large excess of neutrons or protons forces nucleons into higher energy states, reducing the binding energy.

5. **Pairing Term (δ):** Nucleons have a strong tendency to couple in pairs (spin-up with spin-down). This term dictates the **fissile** nature of an isotope:

$$\delta(A,Z) = \begin{cases} +\Delta & \text{for Even-Even nuclei (Most Stable)} \\ 0 & \text{for Odd-Mass nuclei (e.g., } ^{235}\text{U, } ^{239}\text{Pu)} \\ -\Delta & \text{for Odd-Odd nuclei (Least Stable)} \end{cases} \tag{2}$$

It is this **Pairing Term** that creates the "odd-even" effect, explaining why adding a neutron to an odd nucleus (like ^{235}U) releases more energy than adding one to an even nucleus (like ^{238}U).

5 Fissile vs. Fissionable

Now that we understand the Liquid Drop mechanism and the Semi-Empirical Mass Formula (SEMF), we can use these tools to explain the physical difference between **fissile** and **fissionable** isotopes. The distinction lies in the **Pairing Energy** term (δ) of the nuclear binding energy.

When a nucleus captures a neutron, the excitation energy (E_{ex}) of the resulting compound nucleus comes from two sources:

$$E_{ex} = \underbrace{S_n}_{\text{Separation Energy}} + \underbrace{E_k}_{\text{Neutron Kinetic Energy}} \tag{3}$$

For fission to occur, this excitation energy must exceed the critical fission barrier (E_{crit}).

- **Fissile Isotopes (e.g., ^{235}U, ^{239}Pu):**
  These isotopes have an **Odd** number of neutrons. When they absorb a neutron, they become **Even-Even** nuclei. This formation releases a massive "Pairing Energy" bonus, resulting in a very high Neutron Separation Energy (S_n).

  For ^{235}U, $S_n \approx 6.5$ MeV, which is **greater** than the fission barrier ($E_{crit} \approx 5.9$ MeV).

  Result: The neutron needs zero kinetic energy ($E_k \approx 0$). Fission occurs with **thermal** neutrons.

- **Fissionable (Threshold) Isotopes (e.g., ^{238}U):**
  These isotopes already have an **Even** number of neutrons. When they absorb a neutron, they become **Even-Odd** nuclei. They do *not* receive the pairing bonus.

  For ^{238}U, the separation energy is only $S_n \approx 4.8$ MeV, which is **lower** than the barrier ($E_{crit} \approx 5.9$ MeV). The drop vibrates, but does not break.

  Result: To overcome the barrier, the incoming neutron must supply the deficit via kinetic energy ($E_k > 1$ MeV). Fission is a **threshold** reaction.

Summary Definition: A *fissile* isotope fissions with thermal neutrons (due to pairing energy), while a *fissionable* isotope requires a high-energy "kick" to split. Note that all fissile isotopes are fissionable, but not all fissionable isotopes are fissile.

Table 1: Thermal Neutron Fission Yield (ν) for Common Fissile Isotopes

Isotope	**Symbol**	**Origin**	**Thermal ν**
Uranium-233	^{233}U	Bred from Thorium (^{232}Th)	2.49
Uranium-235	^{235}U	Natural (0.7% of Natural U)	2.42
Plutonium-239	^{239}Pu	Bred from Uranium (^{238}U)	2.87
Plutonium-241	^{241}Pu	Bred from Plutonium (^{240}Pu)	2.93

Table 2: Important Threshold Fissionable (Fertile) Isotopes

Isotope	**Threshold**	**Role in Reactor**
Uranium-238 (^{238}U)	~ 1.1 MeV	Dominant fuel component (95%+); breeds ^{239}Pu via capture.
Thorium-232 (^{232}Th)	~ 1.3 MeV	Foundation of the Thorium cycle; breeds ^{233}U via capture.
Plutonium-240 (^{240}Pu)	~ 0.6 MeV	Accumulates in high-burnup fuel; breeds ^{241}Pu. Note the lower threshold.
Uranium-236 (^{236}U)	~ 0.9 MeV	Parasitic buildup from ^{235}U capture; penalizes neutron economy.

Practical Implications: Fast Fission and Fertility

While threshold isotopes cannot sustain a chain reaction on their own (since most fission neutrons are quickly slowed below 1 MeV), they still play two critical roles in reactor physics:

1. **The Fast Fission Bonus (ε):** Fission neutrons are born at high energies (average ≈ 2 MeV). Before they slow down, they have a chance to strike a ^{238}U nucleus and cause fission. In a typical Light Water Reactor, this accounts for 3–5% of all fissions, providing a small but valuable multiplier to the neutron population.

2. **Fertile vs. Parasitic:** Most threshold isotopes are **Fertile**, meaning their capture product leads to a new fissile fuel (e.g., $^{238}\text{U} \rightarrow ^{239}\text{Pu}$). However, some, like ^{236}U, are purely **Parasitic**. ^{236}U is formed when ^{235}U captures a neutron without fissioning. It acts as a neutron poison, absorbing neutrons to become the long-lived waste product ^{237}Np, without contributing to breeding useful fuel.

6 The Products of Fission

Fission is rarely symmetric. The nucleus does not split into two equal halves ($A \approx 118$). Instead, quantum mechanical shell effects cause it to split asymmetrically.

6.1 The Mass Yield Curve

If we plot the probability of producing a fragment of mass A, we see a distinctive "double-humped" curve (the Camel Plot).

- **Light Peak:** Centered around $A \approx 95$ (e.g., Strontium, Krypton).
- **Heavy Peak:** Centered around $A \approx 140$ (e.g., Xenon, Cesium, Iodine).

- **The Valley:** Symmetric fission ($A \approx 118$) is very rare (probability drop of ~600x).

6.2 Why are they Radioactive?

The heavy nucleus (^{235}U) has a huge neutron-to-proton ratio (~1.55) to overcome Coulomb repulsion. Stable light elements ($A \approx 100$) prefer a ratio closer to 1.3.

- When the nucleus splits, the fragments inherit the "parent's" neutron-heavy ratio even accounting for the losses from scission and evaporation.
- They are **Neutron Rich** (far too many neutrons for stability).
- To stabilize, they must convert neutrons into protons via **Beta Decay** (β^-). This is the source of the intense radioactivity of nuclear waste.

7 The Energy Balance (Where is the heat?)

We often quote the energy release of fission as **200 MeV**. As engineers, we need to know exactly where this energy is deposited to design our heat transfer systems.

Energy Form	Approx. Energy	Range	Recoverable?
1. KE of Fission Fragments	168 MeV	$\sim 10\mu$m	**Yes** (Fuel)
2. KE of Neutrons	5 MeV	cm to m	**Yes** (Moderator)
3. Prompt Gamma Rays	7 MeV	m	**Yes** (Shield/Structure)
4. Decay Heat (Betas/Gammas)	~15–20 MeV	N/A	**Yes** (Delayed)
5. Neutrinos	10–12 MeV	∞	**No** (Lost)
Total Recoverable Heat	$\approx$ **190–200 MeV**		

Table 3: Distribution of Fission Energy

Key Engineering Insights:

- **The Fuel Gets Hot:** Over 80% of the energy (Item 1) is Kinetic Energy of the heavy fragments. Because these are highly charged, they interact strongly with matter and stop within micrometers. *All this heat is generated inside the fuel pellet.*
- **The Moderator Gets Warm:** The neutron kinetic energy (Item 2) is deposited in the water/graphite as they slow down.
- **Safety:** Item 4 (Decay Heat) is generated *after* the fission event, over seconds, days, and years. Even if the reactor is shut down, this heat source remains. This is the driver for all meltdown scenarios (e.g., Fukushima).

8 Prompt vs. Delayed Neutrons

While 99.35% of neutrons are "Prompt" (emitted at the moment of fission), a tiny fraction ($\beta \approx 0.0065$) are "Delayed."

- **Origin:** Certain fission products (precursors) undergo beta decay and *then* emit a neutron seconds later.

- **Significance:** If all neutrons were prompt, the reactor would respond to changes in 10^{-4} seconds—too fast to control. The delayed neutrons increase the average "generation time" to effectively seconds, allowing human/mechanical control.

9 Spontaneous Fission: Tunneling Through the Barrier

Until now, we have treated fission as a "threshold" reaction requiring energy input (e.g., neutron capture). However, for very heavy nuclei, fission can occur spontaneously as a form of radioactive decay.

9.1 The Physics of the Phenomenon

Recall the **Liquid Drop Model**. For a nucleus to fission, it must deform from a sphere into a "peanut" shape. This deformation requires energy to overcome the surface tension—this is the **Fission Barrier** ($E_b \approx 5$–6 MeV for Uranium).

In *induced* fission, the binding energy of the incoming neutron provides the excitation to climb *over* this barrier. In *Spontaneous Fission (SF)*, the nucleus remains in its ground state but undergoes **Quantum Tunneling** *through* the barrier.

- Like alpha decay, this is a probabilistic process.
- The probability depends exponentially on the height and width of the barrier.
- As Z^2/A (the fissility parameter) increases, the barrier lowers, and SF becomes more frequent.

9.2 Application 1: Fission Track Dating (^{238}U)

While ^{238}U primarily decays via alpha emission ($T_{1/2} \approx 4.5 \times 10^9$ y), it has a small probability of undergoing spontaneous fission ($T_{1/2,SF} \approx 8 \times 10^{15}$ y).

When a uranium atom inside a mineral (like Zircon or Apatite) fissions, the two heavy fragments fly apart with ~ 170 MeV of kinetic energy. These highly charged heavy ions tear through the crystal lattice, leaving a trail of atomic-scale damage called a **Fission Track**.

- **Etching:** By polishing the mineral and applying acid, these damaged tracks etch faster than the healthy crystal, becoming visible under a microscope.
- **Dating:** The density of tracks is proportional to the age of the rock and its uranium concentration. This technique allows geologists to date cooling events from millions to billions of years ago.

9.3 Application 2: The Weapons Problem (^{240}Pu)

In nuclear weapons design, spontaneous fission is a nuisance. It acts as a source of background neutrons that can trigger a chain reaction prematurely.

When producing Plutonium in a reactor ($^{238}\text{U} + n \rightarrow^{239} \text{Pu}$), side reactions inevitably produce the heavier isotope **Plutonium-240**.

$$^{239}\text{Pu} + n \rightarrow^{240} \text{Pu}$$

Table 4: Spontaneous fission rates of weapons materials. Note the massive difference between Pu isotopes.

Isotope	**SF Half-Life [years]**	**Neutrons per gram-second**
^{235}U	1.0×10^{19}	3×10^{-4}
^{238}U	8.2×10^{15}	0.01
^{239}Pu	8.0×10^{15}	0.02
^{240}Pu	$\mathbf{1.1 \times 10^{11}}$	**920.00**

The "Fizzle" Danger: If a nuclear weapon core contains too much ^{240}Pu, the high background of neutrons will initiate the chain reaction *before* the core is fully assembled or compressed. The energy release will blow the core apart before significant fissioning occurs, resulting in a low-yield **"Fizzle."**

Design Consequence:

- **Gun-Type Weapons (e.g., Little Boy):** Work for Uranium because ^{235}U has negligible SF. They are too slow for Plutonium.
- **Implosion Weapons (e.g., Fat Man):** Required for Plutonium. High explosives must compress the core to criticality so quickly (microseconds) that the probability of a ^{240}Pu neutron firing during assembly is minimized.

This is why "Weapons Grade" Plutonium is defined as having $< 6\%$ ^{240}Pu content.

References

1. Lamarsh, J. R., & Baratta, A. J. *Introduction to Nuclear Engineering*, 4th Ed. Pearson, 2017. (Section 3.7 & 2.12).

2. Bethe, H. A., & Bacher, R. F. "Nuclear Physics A. Stationary States of Nuclei." *Rev. Mod. Phys.* **8**, 82 (1936). `https://journals.aps.org/rmp/abstract/10.1103/RevModPhys.8.82`

3. Bohr, N., & Wheeler, J. A. "The Mechanism of Nuclear Fission." *Phys. Rev.* **56**, 426 (1939).

4. Wikepedia authors: *Spontaneous Fission*. A nice article on spontaneous fission with links to further information.

1 Technical Addendum: The Semi-Empirical Mass Formula Calculation

The total binding energy $B(A, Z)$ is given by:

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_{sym} \frac{(A-2Z)^2}{A} + \delta \tag{4}$$

Where the pairing term δ is:

$$\delta = \begin{cases} +a_p A^{-1/2} & \text{Even-Even (Most Stable)} \\ 0 & \text{Odd Mass (A is odd)} \\ -a_p A^{-1/2} & \text{Odd-Odd (Least Stable)} \end{cases}$$

2 Constants Used

Term	**Symbol**	**Value**	**Note**
Volume	a_v	15.80 MeV	
Surface	a_s	18.30 MeV	
Coulomb	a_c	0.714 MeV	
Asymmetry	a_{sym}	23.20 MeV	
Pairing	a_p	12.00 MeV	Note: Signs vary by text

3 Example 1: Iron-56 (^{56}Fe)

A medium-mass, tightly bound nucleus.

- $A = 56, Z = 26, N = 30$
- Pairing: Even Z, Even N $\rightarrow$ **Positive** δ

Term	**Calculation**	**Energy (MeV)**
Volume (+)	15.8×56	+884.80
Surface (-)	$18.3 \times (56)^{2/3}$	-268.02
Coulomb (-)	$0.714 \times \frac{26(25)}{(56)^{1/3}}$	-121.31
Asymmetry (-)	$23.2 \times \frac{(4)^2}{56}$	-6.63
Pairing (+)	$+12.0 \times (56)^{-1/2}$	+1.60
Total B		**490.44 MeV**

Accuracy: The experimental value is $\approx$ 492.25 MeV.
The error is roughly **0.37%**.

4 Example 2: Uranium-235 (^{235}U)

A heavy, fissile nucleus. Note the massive increase in Coulomb repulsion.

- $A = 235, Z = 92, N = 143$
- Pairing: Even Z, Odd N → **Zero** δ (Odd-Mass)

Term	**Calculation**	**Energy (MeV)**
Volume (+)	15.8×235	**+3713.00**
Surface (-)	$18.3 \times (235)^{2/3}$	-696.88
Coulomb (-)	$0.714 \times \frac{92(91)}{(235)^{1/3}}$	**-966.95**
Asymmetry (-)	$23.2 \times \frac{(235-184)^2}{235} = 23.2 \times \frac{2601}{235}$	-256.78
Pairing (0)	Odd Mass Nucleus	0.00
Total B		**1792.39 MeV**

Analysis of U-235

- **Total Binding Energy (B):** 1792.39 MeV
- **Binding Energy Per Nucleon (B/A):** $1792.39/235 \approx \mathbf{7.63}$ MeV

Comparison to Experiment:
Actual measured binding energy for U-235 is ≈ 1783.87 MeV.

$$\text{Error} = \left| \frac{1792.39 - 1783.87}{1783.87} \right| \times 100 \approx \mathbf{0.48}\%$$

5 Discussion

Notice the shift in dominance between the two examples:

- In Iron-56, the Coulomb term is modest (-121 MeV).
- In Uranium-235, the Coulomb repulsion is massive (-966 MeV). This huge instability caused by proton repulsion is exactly why heavy nuclei are prone to alpha decay and fission.
- The Asymmetry term also jumps significantly for Uranium because it has a large neutron excess ($N = 143$ vs $Z = 92$).